Varun Vijay

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Technical Skills

Embedded Systems: STM32, ARM Cortex-M, FreeRTOS, PWM, GPIO, Timers, Interrupts, UART, SPI, I2C, CAN, ARINC 429, ROS2
Programming: C, C++, Embedded C, Python, SystemVerilog, Verilog, Bash, RISC-V Assembly
Hardware & Debugging: Hardware/Firmware Bring-up, Device Drivers, Oscilloscope, Logic Analyzer, Digital Multimeter, HIL/SIL Testing, Signal Analysis
Development Tools: Git, CMake, Docker, OpenOCD, ST-Link/SWD, STM32CubeIDE, PlatformIO, GTKWave, Icarus Verilog, Ubuntu Linux

Experience

Palitronica Inc. Sep 2026 – Present
Embedded Firmware Developer (Co-op) *Waterloo, ON*

- Architecting a **protocol-agnostic STM32 firmware framework** for PowerFuzzer trigger hardware, replacing protocol-specific implementations with a shared architecture that translates common fuzzing and control commands into **CAN, ARINC 429, UART, and additional embedded interfaces**.
- Developing reusable **protocol adapters and board interfaces** that separate shared triggering, timing, and control logic from transport-specific behavior, reducing duplicated firmware and simplifying integration of new protocols and hardware targets.
- Building a portable **CMake, Docker, and Python CLI workflow** for reproducible firmware development on **x86_64 Linux**, standardizing build, flash, monitor, test, and board-configuration workflows across trigger-board targets.
- Implementing **firmware bring-up, SIL/HIL validation, and root-cause debugging** using OpenOCD, ST-Link/SWD, serial monitoring, and laboratory instrumentation across MCU firmware, communication buses, and board-level electronics.
- Contributing to **PowerFuzzer**, a black-box firmware assurance platform combining automated protocol fuzzing with **power side-channel analysis** to identify anomalous and undocumented embedded-system behavior.

Waterloo Aerial Robotics Group Sep 2026 – Present
Embedded Flight Software Developer *Waterloo, ON*

- Implementing **embedded C firmware for the PineCAN Single Servo Driver Rev 2 target** on an **STM32L431KCU6**, integrating application logic with WARG's lightweight **DroneCAN (UAVCAN v0)** stack for CAN-based UAV peripherals.
- Developing and validating **servo-control, CAN message-handling, and STM32 peripheral functionality**, working with HAL drivers, GPIO, timers, and hardware interfaces for reliable actuator control.
- Contributing to WARG's embedded flight-software stack through **firmware bring-up, driver integration, hardware validation, debugging, and Git-based code review** for autonomous UAV systems.

Real-time Embedded Software Group, University of Waterloo Sep 2025 – Dec 2025
Robotics Research Assistant *Waterloo, ON*

- Integrated an **ARM Cortex-M4 STM32F401 MCU** with a Raspberry Pi-based robotic platform, developing embedded firmware for **PWM actuation, quadrature encoder feedback, and real-time motor control**.
- Performed **hardware/firmware bring-up and system-level debugging** using oscilloscopes and function generators, isolating timing, communication, electronics, and ROS-based software faults.

Acesoft Inc. May 2025 – Aug 2025
Software & Systems Developer (Co-op) *London, ON*

- Developed **Python test automation utilities** to reproduce faults, automate functional validation, and improve repeatability of regression testing across integrated software systems.
- Performed **system and integration testing**, analyzing data flow, API communication, error conditions, and component interactions to identify and isolate root causes of software failures.

Projects

NEXA – FPGA Computer & Hardware Diagnostic Platform | *SystemVerilog, FPGA, GTKWave, Git*

- Designed and implemented a custom **16-bit processor architecture** in SystemVerilog with an original ISA, ALU, register file, program counter, instruction decoder, and control unit.
- Verified processor behavior using **Icarus Verilog and GTKWave** and extended the design with **memory-mapped I/O peripherals** for PWM, sensing, and board-level diagnostics.

Glitch Goblin – Embedded Hardware Validation Platform | *STM32, ESP32, C/C++, Python, Git*

- Developed an **interrupt-driven UART driver** on an ARM Cortex-M4 STM32F401 using hardware interrupts and a ring buffer for non-blocking asynchronous communication.
- Designed the **SERUM binary communication protocol** with framed packets, CRC-16-CCITT error detection, ACK/NACK recovery, sequence tracking, and duplicate-message suppression.
- Integrated PC, STM32, and ESP32 hardware to validate **embedded communication, telemetry, and protocol fault-handling** across a multi-device hardware test platform.

Education

University of Waterloo Sep 2024 – Jun 2029
Bachelor of Applied Science in Computer Engineering | Presidential Scholarship of Distinction *Waterloo, ON*

- **Relevant Coursework:** Digital Circuits and Systems, Electronic Circuits, Embedded Microprocessor Systems, Signals and Systems, Digital Computers, Systems Programming and Concurrency, Algorithms and Data Structures